DO YOU KNOW?

The Iron Pillar – Delhi:

The Iron pillar, also known as Ashoka Pillar, is 23 feet 8 inches high, 16 inches wide and weighs over 6000 kg.



The surprise comes in knowing its age, some 1600 years old,an iron column should have turned into a pile of dust long ago. Despite that, it has avoided corrosion for over the last 1600 years and stands as an evidence of the exquisite skills and knowledge of ancient Indian iron smiths.

A protective film was created through a complicated combination of the presence of raw and unreduced iron in the pillarand cycles of the weather, which helped to create a thin, uniform layer ofmisawiteon the pillar. Misawite is a compound of iron, oxygen and hydrogen which does not rust and gives corrosion resistance.

Summary

- Metallurgy relate to the science and technology of metals.
- A naturally occurring substance obtained by mining which contains the metal in free state or in the form of compounds like oxides, sulphides etc... is called a **mineral**.
- minerals that contains a high percentage of metal, from which it can be extracted conveniently and economically are called **ores**.
- The extraction of a metal of interest from its ore consists of the following metallurgical processes.
  (i) concentration of the ore
  (ii) extraction of crude metal
  (iii) refining of crude metal
- The extraction of crude metals from the concentrated ores is carried out in two steps namely, (i) conversion of the ore into oxides of the metal of interest and (ii) reduction of the metal oxides to elemental metals.
- The graphical representation of variation of the standard Gibbs free energy of reaction for the formation of various metal oxides with temperature is called Ellingham diagram
- Ellingham diagram helps us to select a suitable reducing agent and appropriate temperature range for reduction.

XII U1 Metallurgy - Jerald.indd 19 3/29/2019 6:34:06 PM